

FAREAL AHMED

Full Stack Software Engineer · AI & LLM Systems
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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 4+ years of commercial experience across production SaaS, AI-augmented systems, and technical training. Core stack is Java, Kotlin, and React/TypeScript — with hands-on production work in RAG pipelines, LLM tool-calling, and vector search. Based in Melbourne, open to hybrid and remote.

TECHNICAL SKILLS

Languages & Frameworks: Java, Kotlin, Spring Boot, TypeScript, JavaScript, React, Next.js, Node.js, C#, .NET
AI & LLM: LangChain4j, Ollama, pgvector, RAG pipelines, Claude API, Groq API, Gemini API, MCP (Model Context Protocol), prompt engineering, vector embeddings, semantic search
Databases: PostgreSQL (pgvector, pg_trgm), Microsoft SQL Server, MongoDB, SQLite, Prisma, Hibernate
APIs & Integration: RESTful APIs, GraphQL, OpenAPI/Swagger, WebSockets, MCP servers, webhook orchestration
Cloud & DevOps: Docker, AWS, GCP, DigitalOcean, CI/CD (GitLab), Git, Sentry, Umami, Kibana, Grafana
Practices: Agile/Scrum, TDD, Clean Architecture, Microservices, RESTful API Design

EXPERIENCE

Software Engineer · *Compass Operations Pty Ltd* | Melbourne, AU, Jun 2022 – Sep 2025
Griddle — school timetabling & daily schedule management SaaS · Schools across Australia, UK & Europe

- Delivered full-stack features across timetabling, course selection, exams, and daily organisation for Griddle — a scheduling platform used by hundreds of schools across Australia, UK, and Europe — using Java, Kotlin, Spring Boot, React, and TypeScript.
- Re-engineered an asynchronous job engine into a hybrid sync/async processor using the Factory pattern and Java Suppliers, enabling real-time exam changes to propagate instantly across timetables and daily organisation while preserving 100% code reuse.
- Cut average page load time by 40% through lean DTO design, efficient data-fetching strategies, and targeted query optimisation across multiple React views.
- Resolved 30+ scheduling conflict edge cases via a recursive backtracking algorithm, ensuring 100% data integrity during peak course selection periods.
- Built scalable backend services with RESTful and GraphQL APIs; developed modular, accessible React/TypeScript UIs aligned to enterprise design systems.
- Maintained TDD test suites and monitored GitLab CI/CD pipelines to ensure stable, on-time releases aligned to academic calendar deadlines.
- Mentored junior engineers and led post-incident reviews, contributing to improved code quality standards and faster production incident resolution.

Technical Trainer · *BASIS Institute of Technology & Management* | Dhaka, BD, Feb 2013 – Oct 2014

- Trained 3 cohorts in C#, .NET, OOP, and clean coding practices; designed curriculum based on industry standards.

Software Engineer · *ADDIE Soft Ltd* | Dhaka, BD, Aug 2011 – Jan 2013

- Built core features for Oddhoyon, a school management system used by 1,000+ schools across Bangladesh.
- Led migration from desktop (Windows Forms) to ASP.NET MVC, improving scalability and maintainability.

PROJECTS

TrolleyToday — *AI Grocery Price Comparison PWA* trolleytoday.com

- Mobile-first PWA letting Australian shoppers compare grocery prices across Woolworths, Coles, IGA, Aldi, and specialty grocers — with price-per-unit normalisation (\$/kg, \$/L) and cheapest/most-expensive auto-highlighting.
- Agentic AI assistant using tool-calling architecture: autonomously decides between semantic product search, batch pricing, or a two-step recipe pipeline (LLM ingredient extraction → parallel vector search). The primary model is Llama 3.3-70b via Groq with automatic fallback to Gemini 2.0 Flash on rate-limit.
- Hybrid search combining LLM query expansion, local BGE-small-en-v1.5 embeddings via Xenova/Transformers, pgvector, and pg_trgm fuzzy matching — responses stream token-by-token via ReadableStream.
- Offline-capable shopping list (Zustand + localStorage), quantity steppers, two view modes, and one-tap PDF export grouped by store with subtotals.
- Deployed on DigitalOcean with Docker-managed PostgreSQL 16, IP-based rate limiting, Sentry error tracking, and self-hosted Umami analytics.
- Stack: Next.js 14, TypeScript (strict), Tailwind CSS, PostgreSQL 16, Groq, Gemini, Xenova/Transformers, jsPDF, Sentry, Docker, DigitalOcean

Doc Whisperer — *AI Document Chat — RAG Application* github.com/farealahmed/doc-whisperer

- Full-stack RAG app that lets users upload PDFs and DOCX files and query them via natural language — backend in Spring Boot + Java 21, LangChain4j for retrieval orchestration, Apache Tika for document parsing.
- Text chunked, embedded with all-MiniLM-L6-v2 (384-dim vectors), and stored in PostgreSQL via pgvector for cosine similarity search. Local LLM inference via Ollama (llama3) — all data stays on-machine.
- Frontend in React 18, TypeScript, Tailwind CSS, and Shadcn/UI; full OpenAPI/Swagger documentation across every endpoint.

LearnEasy — *Full-stack LMS on Cloudflare Pages* courses.farealahmed.com

- Full-stack LMS built entirely on the Cloudflare edge: React Router v7 (SSR), D1 for data, KV-backed sessions, R2-hosted HLS video with presigned manifests and magic-link auth.
- Test suite runs with Vitest inside the Workers runtime — no Node.js shims, tests run in the actual execution environment.

NS Sunshine Psychology — *Client Website* nssunshinepsychology.com

- Production client site in React, TypeScript, and Tailwind CSS — live at nssunshinepsychology.com.

EDUCATION

BSc in Computer Science and Engineering 2007 – 2011

Ahsanullah University of Science and Technology, Dhaka, Bangladesh